Karthikeyan Umesh

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Summary

- Cloud & Software Developer with an M.Eng. in Software Engineering (Concordia, 2026); AWS Certified Cloud Practitioner.
- Ships end to end: Terraform-provisioned AWS infrastructure with CI/CD, Python and Java backends, and open-source on-device AI.
- Bilingual (English/French); open to software, cloud/DevOps, AI automation and data roles, and to relocation across Canada.

Skills

- **Languages**: Python, Java, SQL, JavaScript, HTML/CSS
- **Cloud & DevOps**: AWS (EC2, S3, RDS, VPC, IAM, ALB, Auto Scaling, CloudWatch), Terraform, Docker, GitHub Actions, Jenkins, Linux
- **Backend & APIs**: FastAPI, Pydantic, SQLAlchemy, HTTPX, asyncio, Flask, Play Framework, Akka, REST APIs, PostgreSQL
- **AI & Data**: ONNX Runtime, AMD Ryzen AI (NPU), INT8 Quantization, YOLOv8, OpenCV, NumPy, Gemini API, Regression Models
- **Quality & Practices**: pytest, Manual & Automated Testing, Code Reviews, Sonar, Checkov, Object-Oriented Design, Agile/Scrum, Git

Projects

AWS 3-Tier Architecture with Terraform — Terraform, VPC, ALB, EC2 Auto Scaling, RDS, S3, IAM, FastAPI, Docker **GitHub**
- Built a request tracker on a 3-tier AWS stack: VPC across two AZs, ALB, EC2 Auto Scaling running FastAPI, RDS PostgreSQL, S3.
- Applied least-privilege IAM and chained security groups (ALB → app → database); no SSH or public IPs, admin access via SSM.
- Provisioned it all as Terraform modules with remote state; PRs run tflint, Checkov and plan in GitHub Actions via OIDC.
- Tested self-healing: on a failed health check, Auto Scaling replaced the instance and a CloudWatch alarm emailed via SNS.

Sloucher: On-Device AI Posture Coach — Python, ONNX Runtime, AMD Ryzen AI (NPU), YOLOv8n-pose (INT8), OpenCV **GitHub**
- Released an open-source Windows app running AI on the laptop NPU: 96.5% of operators on the NPU at 12–17 ms per frame.
- Cut whole-app CPU use from 44% to 0.8% by tracing ONNX Runtime thread spinning and measuring the full process tree.
- Shipped two releases in 8 days: installer with self-test, checksummed GitHub Releases, tests on every PR, bilingual website.

Serverless AI Resume Tailor API — Python, FastAPI, Gemini API, Pydantic, HTTPX, Docker **GitHub**
- Built a FastAPI service that scores PDF resumes against job descriptions with Gemini via async HTTPX, returning validated JSON.

TubeLytics: Real-Time Data Platform — Java, Play Framework, Akka, WebSockets, REST APIs **Team of 4**
- Built the rate-limited YouTube API service with automatic key switching on quota limits, controllers, and an Akka actor.

Experience

Client Relations & Operations Specialist (Tele Fundraising) **February 2026 – Present**
Fundraising Direct *Remote, Canada*
- Manage high-volume bilingual (EN/FR) donor outreach for national charities; analyze daily metrics to plan weekly workflows.

AI Automation & Data Analyst (Part-Time) **August 2026 – October 2026**
Boston Soft Design *Remote (US-based)*
- Supported AI automation, data analysis and client acquisition for a provider of CRM software and AI integration services.

Junior Engineer **June 2023 – March 2024**
Inventeron Technologies and Business Solutions LLP *Bengaluru, India*
- Developed and tested Python components and automation scripts on Linux; validated PostgreSQL data integrity before release.
- Ran unit and regression testing, used Git with code reviews in Agile teams, and wrote technical documentation.

Machine Learning and Automation Intern **March 2023 – May 2023**
Inventeron Technologies and Business Solutions LLP *Bengaluru, India*
- Built Python ETL steps cleaning data for the team's predictive maintenance model, which cut equipment downtime by 20%.
- Automated S3-based data processing workflows, cutting manual data handling by 30%.

Education

Concordia University **May 2024 – May 2026**
Master of Engineering in Software Engineering *Montreal, QC, Canada*

New Horizon College of Engineering **August 2019 – June 2023**
Bachelor of Engineering in Computer Science Engineering *Bengaluru, India*

Certifications, Awards and Languages

- **AWS Certified Cloud Practitioner** (Dec 2024) | Solutions Architect – Associate (In Progress) | AWS re/Start (12 weeks)
- Springer publication, ICDSAI 2023 | **Best Paper Award**, WCASET 2022 | **Best Project Pitch Award**, Concordia
- **Languages**: English (native/bilingual), French (MIFI Niveau 7), Hindi, Kannada, Tamil